GOOGLE CLOUD / VERTEX AI

Google ADK LlmAgent · Gemini inference · strict structured responses

- **ParserAgent** — floor_plan_parser
- **CritiqueAgent** — fidelity critic
- **OpeningsAudit** — door verifier
- **EditAgent** — typed operations
- **TactileLayoutAgent** — label relocation

authenticated ADK inference bus

BROWSER / HUMAN

INPUT
Floor-plan upload
PDF · PNG · JPG · WebP · ≤10MB
photo or evacuation map

HUMAN-IN-THE-LOOP
Editable canvas
direct geometry edits
plain-English correction

human canvas reads/writes latest version

BUN PROCESS / HONO API

HTTP 202 · process-local jobs · browser polling · Postgres model versions

INGEST + JOB DISPATCH

HTTP API — **Validate + persist**
MIME · size · source file

DETERMINISTIC — **Raster + crops**
PDF page 1 → PNG
downscale + detail regions

BACKGROUND JOB — **Parse dispatch**
202 + job id · poll status

POSTGRES + FILES — **Project state**
jobs · status · source paths
FloorModel JSON · artifacts
STL metadata + triangle counts

REJECT · refinement context · maximum 5 iterations

SEMANTIC EXTRACTION + CRITIQUE

VERTEX AI — **ParserAgent**
IN plan + crops + dimensions
OUT strict FloorModel JSON

DETERMINISTIC — **Normalize + audit**
IDs · topology · ink coverage
source overlay render

VERTEX AI — **CritiqueAgent**
verdict · findings
confidence · source comparison

PARSE ACCEPT?

ACCEPT

typed operation → validate IDs → append version

DOOR AUDIT + HUMAN REVIEW

VERTEX AI — **OpeningsAuditAgent**
magnified door crops
keep · move · delete · add

APPEND-ONLY — **FloorModel versions**
immutable revision chain
latest accepted model

VERTEX AI — **EditAgent**
model + selection
typed ops or clarification

violations → relocate → revalidate · maximum 4

TACTILE SYNTHESIS + COMPLIANCE

DETERMINISTIC — **Tactile converter**
plate · braille · symbols
legend · relief geometry

RULE ENGINE — **Validate + repair**
scale · margins · clearance
seams · label fit · dot geometry

VERTEX AI — **TactileLayoutAgent**
moves labels/symbols only
never accepts a worse state

ZERO VIOLATIONS?

violations remain · 409

GEOMETRY / EXPORT

EXPORT GATE
Latest run is done + valid
Manifold builds watertight map,
braille plate, and legend meshes

OUTPUT
Print-ready STL bundle